

The Phonology of Biblical Hebrew

Sound Change and Allophony in a Dead Language

Benjamin Suchard

Chinese University of Hong Kong Linguistics Seminar
1 March 2022



Universiteit Leiden

Ancient Hebrew: Israel and Judah, ca. 12th c. BCE–2nd c. CE



Courtesy of the Oriental Institute of the University of Chicago

The Hebrew Bible: religious texts from 1st millennium BCE



Oral transmission of pronunciation

- Hebrew script originally only wrote consonants

בְּנִימִין זֶאֱבַב יִטְרֹף בְּבֹקֶר יֹאכֵל עַד וְלַעֲרֵב יַחֲלֹק שָׁלָל

bnymin zʔb yṭrp bbqr yʔkl šd wlsrb yḥlq šll

- Pronunciation of Hebrew Bible texts fixed around 100 CE

**benyāmīn zeʔēb yeṭrāp; bab-boqr yōkal šad wa-laš-šarb yaḥalleq šālāl.*

‘Benjamin is a rending wolf: in the morning he devours prey and in the evening he divides spoil.’ (Genesis 49:27)

- Centuries of dual transmission: consonantal text together with oral reading tradition

Language of the Masoretic Text: (Tiberian) Biblical Hebrew



The Neogrammarian Hypothesis

Sound change is **phonetically regular**: the same sound in the same phonetic environment in the same language undergoes the same changes.

Proto-(West-)Germanic	English
* <i>kinn-</i>	<i>chin</i>
* <i>kilp-</i>	<i>child</i>
* <i>kirik-</i>	<i>church</i>
* <i>kald-</i>	<i>cold</i>
* <i>krupp-</i>	<i>crop</i>
* <i>kuning-</i>	<i>king</i>

Exceptions to regular sound change: I

Proto-Germanic	English
* <i>kilp-</i>	<i>kilt</i> from Old Norse
* <i>kid-</i>	<i>kid</i> from Old Norse
?	<i>kiwi</i> from Maori



Exceptions to regular sound change: II

- Proto-West-Germanic **sōkijan* > Old English *sēcan* ‘to seek’
- **k* preserved in *sēcð* ‘(he/she/it) seeks’
- *k* analogically reintroduced to the rest of the verb: apparent exception to the sound change
- analogy does not have to be regular (cf. *beseech*)

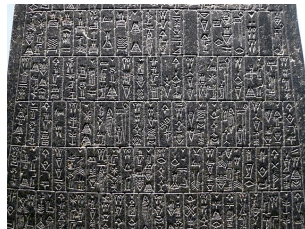
Exclude loanwords + exclude cases of analogy + identify phonetic conditioning: left with regular sound change (sound law)

Do the Biblical Hebrew vowels follow sound laws?



Introducing Hebrew's Semitic relatives

- Aramaic (Syria and Mesopotamia, 1st millennium BCE–present)
- Arabic (Middle East & North Africa, 1st millennium CE–present)
- Akkadian (Mesopotamia, 3rd–1st millennium BCE)



The Canaanite Shift

* $\bar{a} > \bar{o}$

- * $sal\bar{a}m-$ > $\check{s}al\bar{o}m$ ‘wellbeing’; cf. Aramaic $\check{s}l\bar{a}m$
- * $\text{?}ayl-\bar{a}n-$ > $\text{?}\bar{e}l\bar{o}n$ ‘big tree’, cf. Aramaic $\text{?}il\bar{a}n$
- * $yib\bar{a}\theta u$ > $y\bar{e}b\bar{o}\check{s}$ ‘he is put to shame’; cf. Akkadian $ib\bar{a}\check{s}$ ‘he was put to shame’

But sometimes, * $\bar{a} > \bar{a}$:

- * $^ts\bar{a}^ts-$ > $s\bar{a}s$ ‘moth’, cf. Aramaic $s\bar{a}s$
- * $\text{?}an\bar{a}n-$ > $\text{?}\bar{a}n\bar{a}n$ ‘cloud cover’, cf. Arabic $\text{?}an\bar{a}n-$
- * $qiny-\bar{a}n-$ > $qiny\bar{a}n$ ‘possessions’, cf. Aramaic $qiny\bar{a}n$

Only stressed * $\bar{a} > \bar{o}$? Cf.:

$\text{?}\bar{e}n\bar{o}\check{s}$ ‘human’, $\text{?}\bar{a}n\bar{a}\check{s}-\bar{i}m$ ‘men’

$r\bar{o}\check{s}$ ‘head’, $r\bar{a}\check{s}-\bar{i}m$ ‘heads’

?

Getting rid of borrowed and analogical forms

- *sās* ‘moth’: unpredictable consonants and vowels in other languages, cf. Arabic *sūs*- ‘moth worm’, Ge‘ez *ṣāṣē* ‘moth, worm’.
Borrowed, onomatopoeia?
- **-ān*- suffix is very productive; either *-ōn* or *-ān* could have spread to words like *ʔēlōn* ‘big tree’ and *qinyān* ‘possessions’
- *ʕānān* ‘cloud cover’ could be patterned after other *CāCāC* weather words from **CaCaC*-, like:
māṭār ‘rain’ < **maṭar*-, cf. Arabic *maṭar*-
bārāq ‘lightning’ < **baraq*-, cf. Aramaic *braq*

etc.

Many remaining examples didn't have **ā*

- *ʔānāš-īm* 'men' < **ʔinas-īma*: cf. possessed form *ʔanšē* 'men of' (long **ā* would be preserved)
- *rāš-īm* 'heads' < **raʔas-īma*
- Habitual *CaCCāC* nouns and adjectives like *dayyān* 'judge' < **CaCCāC*-, cf. Aramaic *dayyān*?
→ no, **CaCCaC*- like Akkadian *šarraḳ*- 'thief' etc.

What's left?

- No evidence for stress conditioning
- Several native words with $*\bar{a} > \bar{a}$:
 - *krāḥ-ayim* ‘shins’ < $*kur\bar{a}\dot{\mathfrak{h}}-ayma$, cf. Arabic *kurāḥ-*
 - *tōšāḇ* ‘resident alien’ < $*taw\theta\bar{a}b-$, cf. Aramaic *tawtāḇ*
 - *mōrāš-ē* ‘desires of’ < $*mawr\bar{a}\theta-ay$ (no other source for $\bar{a}$)
- In all other native words, $*\bar{a} > \bar{o}$

Sound law: $*\bar{a} > \bar{o}$ except after $*u$ or $*w$.

Exceptions: **ā* > *ō* after **u*

- *brōš* ‘juniper’ < **burāθ*-, cf. Akkadian *burāš*-
- *plōnī* ‘someone’ < **pulān*-īy-, cf. Arabic *fulān*-
- *ptōt*-ē ‘morsels of’ < **putāt*-ay, cf. Arabic *futāt*-
- *rimmōn* ‘pomegranate’ < *rummān*-, cf. Arabic *rummān*-

Earlier sound law (‘*rimmon* rule’): unstressed **u* > **i* next to bilabials.

- 1 **rummān*-
- 2 **rimmān*- (unrounding)
- 3 **rimmōn*- (Canaanite Shift)



Another source of $\bar{a}$: tonic lengthening

- $zāh\bar{a}b$ ‘gold’ < * $\dot{\delta}ahab-$, cf. Aramaic $dhāb$
- but: $kātāb$ ‘he wrote’ < * $katāba$, cf. Aramaic $kṭāb$
- morphological conditioning of sound change?

Tonic lengthening: #notallnouns

- Most nouns and adjectives undergo tonic lengthening, e.g.:
 - *yāḏ* ‘hand’ < **yad*-, cf. Arabic *yad*-
 - *yārāq* ‘green (m.sg.)’ < **warāq*-, cf. Akkadian *waraq*-
 - *ʾōlām* ‘eternity’ < **ʾālām*-, cf. Aramaic *ʾālam*
- But many don’t, e.g.:
 - *śaq* ‘sack’ < **śaqq*-, cf. Aramaic *śaqq-ā*
 - *ʾām* ‘people’ < **ʾamm*-, cf. Aramaic *ʾamm-ā*
 - *qal* ‘light (m.sg.)’ < **qall*-, cf. Akkadian *qall*-
- Most verbs don’t undergo tonic lengthening, e.g.:
 - *kātāb* ‘he wrote’ < **katāba*, cf. Arabic *kataba*
 - *yišmaʿ* ‘he hears’ < **yismaʿu*, cf. Arabic *yismaʿu*
- Verbs with a certain structure do:
 - *ʾākālātām* < **ʾakalat-am* ‘it consumed them’

Conditioned by syllable structure

Nouns and adjectives had case endings: nominative **-um*, genitive **-im*, accusative **-am*

- Lengthening: **ḏahab-um*, **yad-um*, **waraq-um*, **ṣālam-um*, **ʔakalat-am...*
 - No lengthening: **kataba*, **yismaʕu*, **śaqq-um*, **ʕamm-um*, **qall-um*
 - Lengthening in open syllables only. What about the verbs?
-
- 1 Final vowels lost (apocope): **kataba*, **yismaʕu* > **katab*, **yismaʕ* etc.
 - 2 Tonic lengthening in open syllables: **yad-um* > **yād-um* etc.
 - 3 Case endings lost: **yād-um*, **śaqq-um* > **yād*, **śaqq* etc.

Tonic lengthening is phonetically conditioned; Neogrammarian Hypothesis is safe!

- Regular sound changes identified:
 - Canaanite Shift + *rimmon* rule
 - Tonic lengthening
 - Collapse of diphthongs/triphthongs, stress shifts, shifts of short **a* and **i*, loss of final vowels...
- All early changes
- What about the reading tradition?

Phonemes and allophones

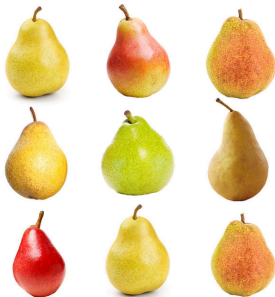
Definition

phonemes: contrastive sounds within a language

Definition

allophones: non-contrastive sounds belonging to the same phoneme

Minimal pairs



[pʰɛʁz]



[bɛʁz]

Conditioned allophony

- *ship* [ʃɪp^h]
- *fit* [fɪt^h]
- *tick* [t^hɪk^h]
- *kiss* [k^hɪs]
- *whiff* [wɪf]
- *nib* [nɪb]
- *bid* [bɪd]
- *rig* [ɹ^wɪg]
- *Liz* [lɪz]
- *live* [lɪv]

Unconditioned allophony



[k^hæt^h], [k^hæt̚], [k^hæʔ]...

Two sounds are phonemically contrastive if & only if their distribution *is* conditioned, but *not* phonetically conditioned

- nothing predicts which sound appears: **allophones**
(like final [t^h], [t̪], [ʔ])
- phonetic environment predicts which sound appears: **allophones**
(like [ɾ] before voiceless obstruents vs. [r] before voiced obstruents)
- anything else predicts which sound appears: **different phonemes**
(like initial [p^h] in some words vs. [b] in others)

Minimal pairs prove two sounds are contrastive, but they are not necessary

Split of *e and *o

- **yahalleq* > *yhallēq* ‘he divides’
- **benyāmīn* > *binyāmīn* ‘Benjamin’
- **ben-yamīnī* > *ben-ymīnī* ‘Benjaminite’

Stressed *e > ē; unstressed *e > i/e

- **koll* > *kōl* ‘all’ (stressed)
- **koll-ō* > *kull-ō* ‘all of it’
- **koll-* > *kol-* ‘all’ (proclitic)

Stressed *o > ō; unstressed *o > u/o

Split of unstressed *e, *o is phonemic

- *e* usually next to pharyngeals/gutturals, *i* elsewhere:

- *ḥedrō* ‘his room’
- *biṭnō* ‘his belly’
- *dimkem* ‘your (m.pl.) blood’

- But not always, e.g.:

- *memšālā* ‘authority’
- *yedkem* ‘your (m.pl.) hand’

- *u* usually before geminates, *o* elsewhere:

- *ḥuqqīm* ‘laws’
- *qodšō* ‘his sanctuary’
- *hoglā* ‘he was exiled’

- But not always, e.g.:

- *šulhān* ‘table’
- *muglīm* ‘exiled (m.pl.)’

Consistent difference, no phonetic conditioning: different phonemes

An unnatural phonemic split

If sound change is phonetically regular, how did **e* and **o* split like this?

Solution:

- [i] ~ [e], [u] ~ [o] originally unstressed allophones of /e/, /o/
- Reading tradition transmitted by speaker whose L1 contrasted unstressed /i/ : /e/ and /u/ : /o/ (maybe Greek)

Teacher thinks...	Teacher says...	Student thinks...	meaning
<i>be</i> <u>ṭ</u> <i>nō</i> /e/	<i>bi</i> <u>ṭ</u> <i>nō</i> /e/	<i>bi</i> <u>ṭ</u> <i>nō</i> /i/	‘his belly’
<i>ye</i> <u>d</u> <i>kem</i> /e/	<i>ye</i> <u>d</u> <i>kem</i> /e/	<i>ye</i> <u>d</u> <i>kem</i> /e/	‘your hand’

Artefact of Biblical Hebrew as a learned (L2) reading tradition

(Near-)minimal pairs with the same meaning

heḡlā nbuzarʔāḏān

‘Nebuzaradan exiled’ (Jeremiah 52:30)

w-hiḡlā ʔet-kol-yrūšālayim

‘And he exiled all of Jerusalem’ (2 Kings 24:14)

wayyīp̄ b-ḡōḏlō

‘And it was beautiful in its greatness’ (Ezekiel 31:7)

halālūhū k-rōḡ gudlō

‘Praise him according to his abundant greatness’ (Psalm 150:2)

kōl ymē ḥayyekā

‘all the days of your life’ (Genesis 3:17)

kol ʔāḥē-rāš šnēʔūhū

‘all a poor man’s brothers hate him’ (Proverbs 19:7)

Does Biblical Hebrew behave like a natural language in terms of phonology?

- Yes: regular sound changes in the living language
- No: allophones unnaturally frozen in the reading tradition

**tawdiy-at-* > תודה **tōdā* > תודה: *tōdā*